CHUCK-A-LUCK SIMULATOR: A STATISTICAL INQUIRY INTO THE LAW OF LARGE NUMBERS AND FREQUENTIST INTERVALS

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1 Introduction

Chuck-a-Luck, also known as Bird Cage, is a game where players bet on a number from 1 to 6. Three fair dice are rolled, and the player is paid based on the number of matching dice. Standard payouts are usually 1:1 for a single match, 2:1 for two matches, and 3:1 for three matches. If no dice match the chosen number, the wager is lost. This game is used to demonstrate fundamental concepts in probability and statistics. This report analyzes a simulation engine developed in R Shiny[1]. The primary objectives are to verify the Law of Large Numbers (LLN) and evaluate Frequentist Confidence Intervals in quantifying uncertainty for winning probabilities.

2 Theoretical Foundations

Chuck-a-Luck is a binomial experiment where a player bets on a number $b \in \{1, 2, ..., 6\}$. Three fair dice are rolled, and the number of matches X follows a Binomial distribution $B(3, 1/6)$. The sample space contains 216 combinations, summarized in Table 1.

Table 1: Chuck-a-Luck Theoretical Outcome Distribution

Outcome (Matches)	Exact Ratio	Probability (%)
0 Matches (Loss)	$125/216$	57.87%
1 Match	$75/216$	34.72%
2 Matches	$15/216$	6.94%
3 Matches	$1/216$	0.46%

The House Edge represents the operator's mathematical advantage, calculated as the negative of the Expected Value $E[X]$ for a \$1 wager:

$$E[X] = \sum_{i=0}^{3} P(X = i) \cdot \text{Payout}_i \tag{1}$$

Using standard payouts ($\{-1, 1, 2, 3\}$), we have:

$$E[X] = \frac{125}{216}(-1) + \frac{75}{216}(1) + \frac{15}{216}(2) + \frac{1}{216}(3) = -\frac{17}{216} \approx -7.87\%$$

This means that for every \$100 wagered, a player is statistically expected to lose \$7.87.

3 Architecture and Implementation

The simulator is implemented using the R programming language R Core Team [2025] and the Shiny web framework Chang et al. [2026].

[1]Source code repository: `https://github.com/qu1r0ra/chuck-a-luck-simulator`.

3.1 Backend Logic and Optimization

To support large-scale simulations, we utilize a vectorized backend. Successes for each trial are calculated using row-weighted sums: `Matches <- rowSums(dice == target)`. This optimization enables sub-second execution even for millions of trials. Additionally, simulation state is persisted across multiple runs using `reactiveValues`, allowing users to cumulatively build a large dataset without reloading the session.

3.2 User Interface Design

The application interface is designed to illustrate both the simulation setup and the subsequent analysis phases. Figure 1 and 2 highlight the transition from a clean initial state to a populated dashboard.

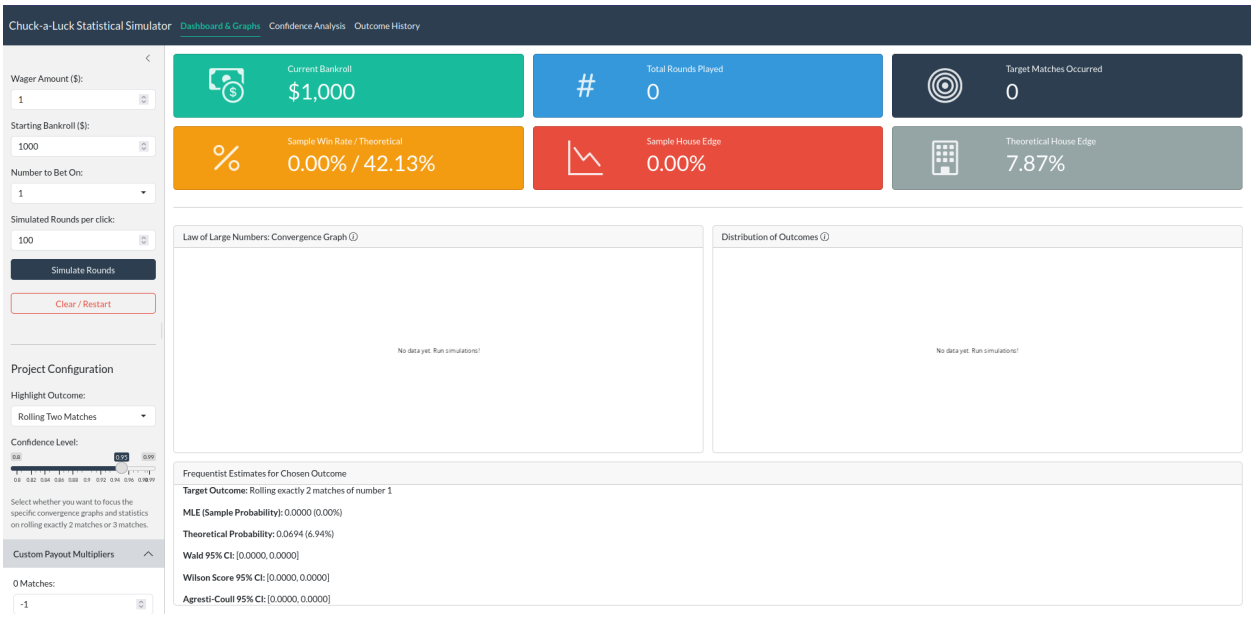

Figure 1: The Chuck-a-Luck Simulator initial setup view.

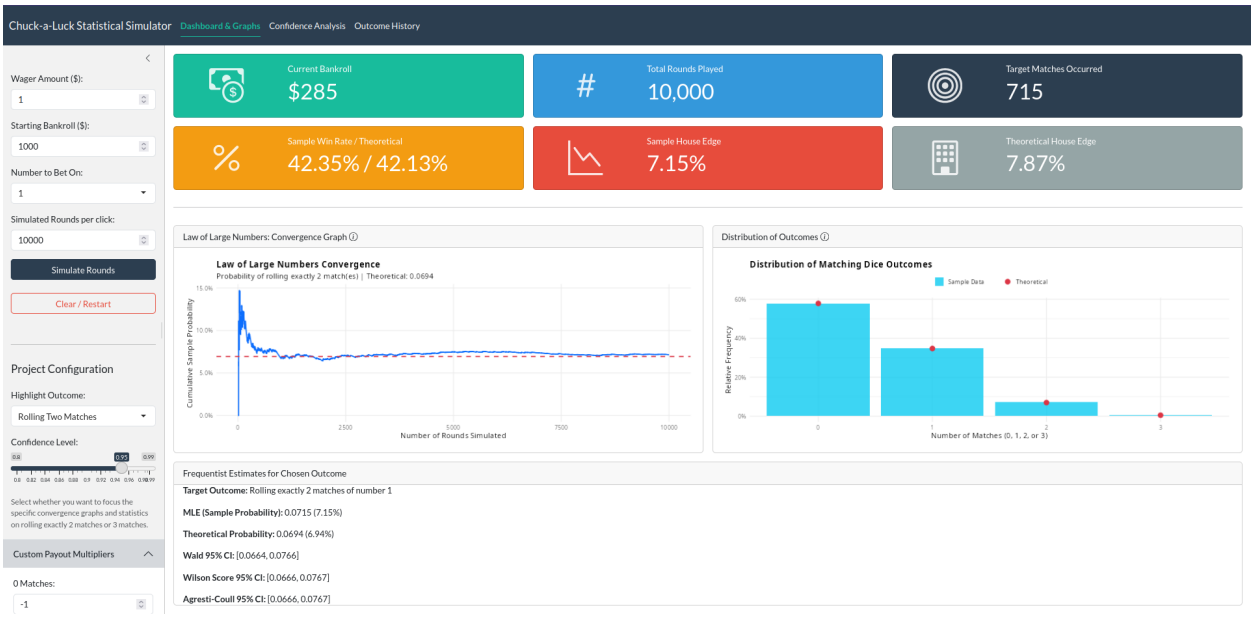

Figure 2: The Simulator interface after $n = 10,000$ trials, showing populated metrics and visualizations.

4 Statistical Analysis

We utilize the simulation engine to demonstrate frequentist convergence and quantify estimator uncertainty.

4.1 Law of Large Numbers and Convergence

As $n \to \infty$, the MLE estimate $\hat{p}$ converges to its theoretical counterpart Taboga [2021]. Figure 3 demonstrates this stabilization across increasing trial counts.

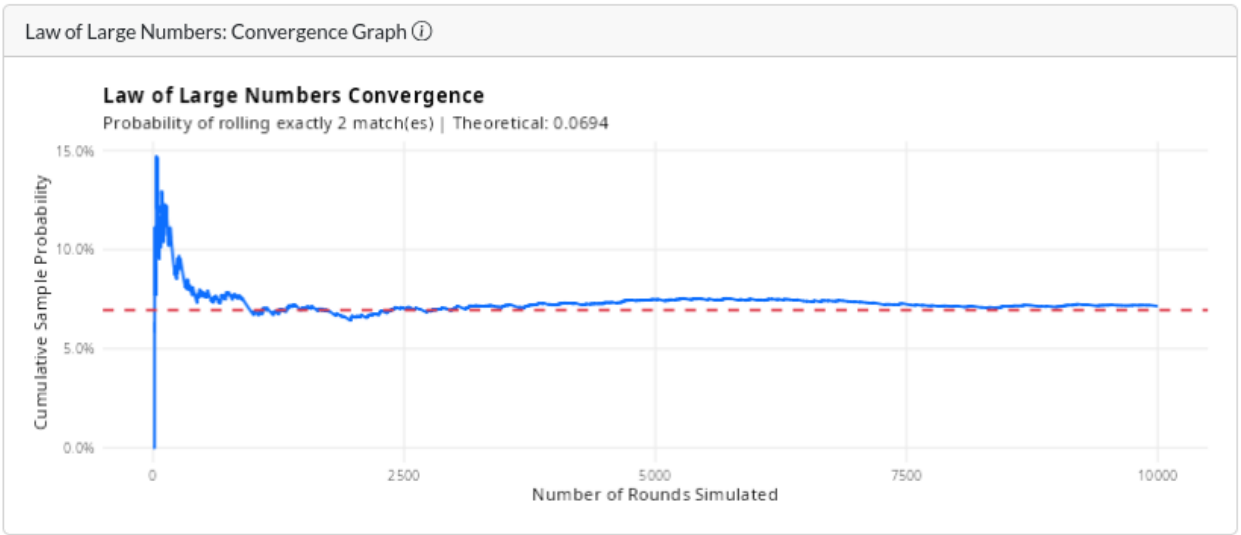

Figure 3: Convergence Trace Plot demonstrating the Law of Large Numbers as the sample proportion stabilizes around $p = 1/6$.

4.2 Frequentist Confidence Intervals

To quantify uncertainty for the estimate $\hat{p}$, we implement Wald, Wilson Score, and Agresti-Coull intervals. Let z be the $1 - \alpha/2$ quantile of the standard normal distribution:

$$\text{Wald:} \quad \hat{p} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \tag{2}$$

$$\text{Wilson Score:} \quad \frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}} \tag{3}$$

$$\text{Agresti-Coull:} \quad \tilde{n} = n + z^2, \quad \tilde{p} = \frac{X + z^2/2}{\tilde{n}}, \quad \tilde{p} \pm z\sqrt{\frac{\tilde{p}(1-\tilde{p})}{\tilde{n}}} \tag{4}$$

Figure 4 presents an integrated view of these three methodologies as implemented in the simulator's analysis dashboard.

Analytical results demonstrate that for rare outcomes at small sample sizes, both the Wald and Agresti-Coull intervals can yield invalid limits below 0, whereas the Wilson Score method remains robustly and naturally bounded within the probability space. As n increases, all three methodologies converge as the binomial distribution approaches normality.

5 AI Integration and Citation

This project was developed with the assistance of Google's Gemini. AI was utilized specifically for:

1. Multi-file architectural planning and repository structure design.

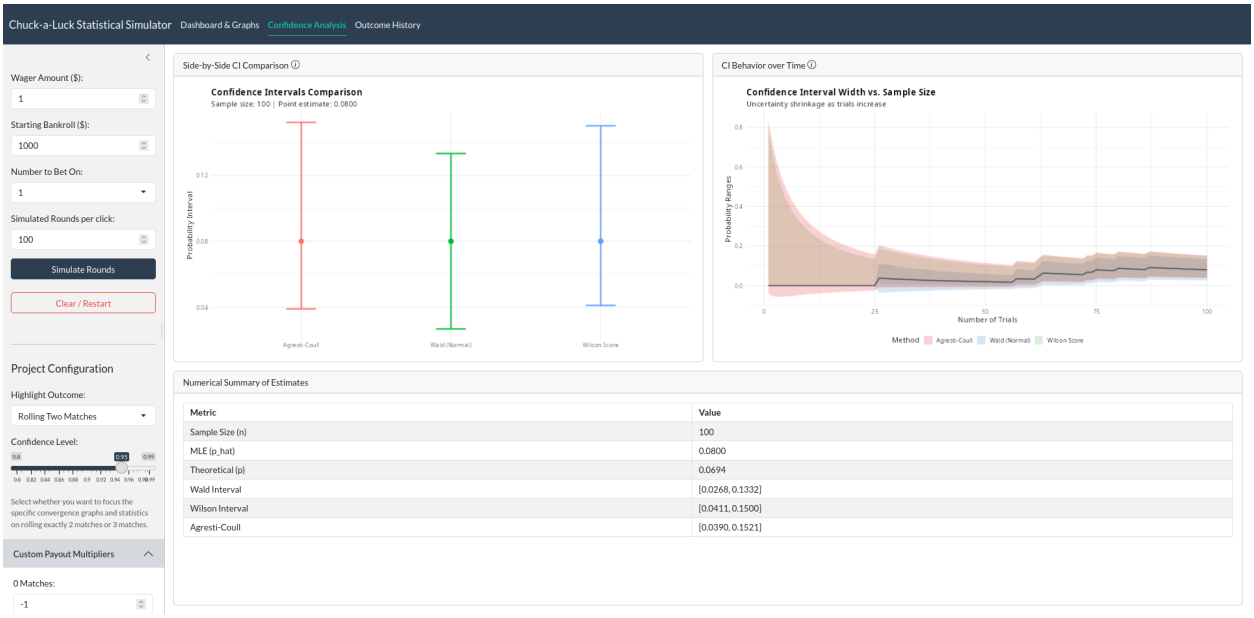

Figure 4: Integrated Confidence Analysis dashboard, showcasing interval methodologies, numerical summaries, and $1/\sqrt{n}$ width shrinkage behavior.

2. Code optimization and refactoring for vectorized simulation performance.
3. Mathematical derivation verification and LaTeX typesetting.
4. Generation of a comprehensive unit testing suite and validation checks.
5. Synchronization of technical documentation across various components.
6. UI/UX layout design and aesthetic refinement of the Shiny application.
7. Debugging complex reactive dependencies and state persistence logic.

References

R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2025. URL `https://www.R-project.org/`.

Winston Chang, Joe Cheng, JJ Allaire, Carson Sievert, Barret Schloerke, Garrick Aden-Buie, Yihui Xie, Jeff Allen, Jonathan McPherson, Alan Dipert, and Barbara Borges. *shiny: Web Application Framework for R*, 2026. URL `https://shiny.posit.co/`. R package version 1.13.0.9000.

Marco Taboga. *Law of Large Numbers, Lectures on probability theory and mathematical statistics*. Kindle Direct Publishing, 2021. URL `https://www.statlect.com/asymptotic-theory/law-of-large-numbers`.